

Article

Ground-Based Cloud Image Segmentation Method Based on Improved U-Net

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Abstract: Cloud image segmentation is a technique that divides images captured by meteorological satellites or ground-based observations into different regions or categories. By extracting the distribution, shape, and dynamic features of clouds, it provides precise data support for the meteorological and environmental fields, significantly influencing photovoltaic (PV) power generation forecasting, astronomical telescope observatory site selection, and weather forecasting. A ground-based cloud image segmentation model based on an improved U-Net is proposed, which adopts an overall encoder–decoder structure. In the encoder phase, this paper constructs a dilated convolution–atrous spatial pyramid pooling (ASPP)–dilated convolution structure to enhance early cloud feature extraction. Dilated convolution is a novel type of convolution that expands the receptive field by inserting holes into standard convolution, thereby capturing a larger range of contextual information. ASPP maintains high resolution while paying attention to both local details and global structures of the image. In the decoder stage, the bicubic interpolation method is used for up-sampling to restore the feature map resolution and improve the clarity of the segmented image. The bicubic interpolation method refers to the use of cubic polynomial functions to interpolate the pixel values of the input image. In addition, this paper designs a novel skip connection layer structure between the encoder and decoder, composed of a depthwise separable path (DS path) and an improved channel spatial attention module (Im-CSAM) connected in sequence. The DS path combines depthwise separable convolutions and residual structures to facilitate information exchange between high-level and low-level features. The Im-CSAM is a modular attention mechanism that focuses on important cloud features in spatial and channel dimensions to enhance segmentation accuracy. Experiments show that compared to the traditional U-Net, the accuracy, precision, and MIoU of this model improved by 2.2%, 4.1%, and 5.0%, respectively, in the SWINySEG dataset, and by 3.2%, 3.6%, and 5.8%, respectively, in the TCDD dataset, proving that the improved method has a better generalization ability and segmentation performance.



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1. Introduction

Clouds are a common and important natural phenomenon consisting of water droplets, ice crystals, or a mixture of both, forming various shapes that float in the sky, covering more than 60% of the total land area on Earth [1]. Clouds play an important role in maintaining the atmospheric radiation budget balance by suppressing both shortwave and longwave solar radiation. Additionally, clouds also play a significant role in Earth's heat balance and water vapor cycle; they not only reflect current atmospheric movements and water vapor changes but also play a crucial role in regulating and indicating future weather and climate changes, making them one of the key factors influencing global climate change [2].

Therefore, accurately obtaining sky information through cloud observation is of positive significance for the fields of meteorology [3] and astronomy [4].

Since ancient times, cloud observation has been a very important and challenging task. With the continuous invention and improvement of various hardware devices and instruments, the method of cloud observation is gradually shifting from manual observation to automated observation. Automated observation is based on imaging devices that simulate the human eye for cloud observation. Currently, automated observation is mainly divided into satellite remote sensing and ground-based remote sensing. Satellite remote sensing primarily uses satellite equipment to obtain multi-band images; ground-based remote sensing mainly uses total sky imagers or panoramic fisheye cameras, and the obtained images are ground-based cloud images. However, satellite cloud observation can only capture information about the tops of clouds, and the description of mid-lower layer clouds or cloud base information is not clear enough, leading to satellite cloud observation currently not meeting the needs of relevant scientific research. In contrast, ground-based cloud images mainly reflect the local area of the sky and the distribution, change, and movement of cloud layers, allowing for effective and accurate acquisition of information about cloud cover, cloud height, and cloud type. Therefore, ground-based observation is an important part of research related to cloud observation.

Cloud image segmentation is the first step in cloud observation. In the fields of computer vision and remote sensing, cloud image segmentation aims to accurately extract cloud regions from Earth observation images and classify each pixel of the image into cloud or non-cloud categories. It is roughly divided into three categories: threshold-based methods, traditional machine learning-based methods, and deep learning methods.

Thresholding is one of the most commonly used and simplest methods in traditional image segmentation. Traditional threshold segmentation is often performed by the relative difference between the red channel and the blue channel in the three RGB channels of an image. Long et al. [5] and Kreuter et al. [6] proposed a fixed threshold algorithm which uses the ratio of red and blue channel values to achieve cloud–sky recognition from cloud images. Similarly, Heinle et al. [7] used the red and blue difference as a fixed threshold, where the segmentation results are better than the red and blue ratio method but still produce a small error. However, considering the complexity of cloud images, such algorithms are not effective in segmenting thin clouds. Yang et al. [8] proposed an adaptive threshold segmentation method based on the OTSU algorithm. By comparing the adaptively calculated threshold with the grayscale value after processing the blue and red bands, the sky and cloud pixels are distinguished. However, cloud cover can severely affect the accuracy of adaptive thresholds. Based on the adaptive threshold method, a ground-based cloud image segmentation method based on local threshold interpolation has been proposed [9]. Li et al. [10] proposed the hybrid thresholding algorithm (HYTA) for ground-based cloud image detection, which takes full advantage of the combination of fixed and adaptive thresholding methods. However, the above methods are greatly affected by factors such as light intensity and imaging quality of the equipment, and lack of robustness. The area around the sun in cloud images is easily misidentified as clouds, leading to excessive segmentation of cloud images and reduced detection accuracy.

In addition to the threshold algorithm, Shi et al. [11] proposed a new ground-based cloud image segmentation method that uses a superpixel-based GM (graph model) to integrate multiple information sources. However, this method may result in over-segmentation and multiple connected regions during the iterative process. Taravat et al. [12] applied a multilayer perceptron (MLP) and a support vector machine (SVM) to ground-based cloud image segmentation for the first time. This method inputs the pixel values of the red, green, and blue bands of the cloud image. After processing with the MLP and SVM, the output will classify the pixels based on whether the area is covered by clouds. The performance of the segmentation is optimized by continuously adjusting the parameters of the MLP and SVM, but the segmentation accuracy needs to be improved. Blazek et al. [13] proposed a segmentation method based on K-means color space. This method first transforms the

color of the cloud image and converts the RGB color space into the CIE Lab color space, thereby separating the photometric and chromatic information. However, the centroid can lead to slow convergence of the algorithm when it is chosen completely at random. The abovementioned traditional machine learning cloud image segmentation methods have improved segmentation quality to a certain extent compared to threshold algorithms. However, such methods are still affected by factors such as light intensity, and the quality of cloud image segmentation needs further improvement.

With the development of deep learning, deep convolutional neural networks have become a hotspot in the field of vision, and more and more deep learning models are applied to cloud image segmentation tasks. Ronneberger et al. [14] proposed U-Net for image segmentation, which effectively integrates shallow detail information and deep semantic information through skip connection layers. Dev et al. [15] used a lightweight network CloudSegNet to simultaneously complete the two tasks of segmenting daytime and nighttime ground cloud images. Shi et al. [16] proposed an automatic segmentation method for ground-based cloud images based on the enhanced fully convolutional neural network (EFCN). In addition, Shi, Zhou, and Qiu [17] proposed a new network Cloud U-Net to segment ground-based cloud images, which performed better than the EFCN.

The categories and technical methods used in the above research and literature related to cloud image segmentation tasks are detailed in Table 1.

Table 1. An introduction to the main categories and methods of ground-based cloud segmentation.

Segmentation Method Category	Reference	Technique	Comment
threshold-based method	[5]	Red–blue ratio method	Dependent on weather conditions and spectral characteristics
	[6]		
	[7]	Red–blue difference method	Difficult to deal with thin clouds
	[8]	OTSU algorithm	Greatly affected by cloud cover
	[9]	Local threshold interpolation	Greatly affected by light intensity and imaging quality of equipment
	[10]	HYTA	Solar halo region is easily misidentified as clouds
traditional machine learning-based method	[11]	Superpixel method	Over-segmentation and multiple connected regions
	[12]	MLP, SVM	Segmentation accuracy needs to be improved
	[13]	K-Means	Algorithm converges slowly
Deep learning method	[14]	U-Net	Better segmentation accuracy, complex network
	[15]	CloudSegNet	Model structure simplification; segmentation accuracy needs to be improved
	[16]	EFCN	Low utilization rate of global information and local information
	[17]	CloudU-Net	Better than EFCN, but segmentation efficiency is not high

However, most of the existing deep learning models ignore the information exchange between low-level and high-level features, and important cloud features in spatial and channel dimensions are undervalued, resulting in poor model training accuracy, poor generalization, and poor segmentation. This paper proposes a cloud image segmentation method that integrates novel convolution and attention mechanisms. By introducing dilated convolutions and depthwise separable convolutions into U-Net, and improving the CBAM attention mechanism, the segmentation accuracy and effectiveness of the model are enhanced. The contributions of this paper are as follows:

- (1) A dilate block is constructed to extract shallow cloud image features while retaining the fine structure of the cloud image, reducing the loss of cloud information, and capturing a larger range of contextual information.
- (2) A depthwise separable residual structure DS path is constructed. By stacking depthwise separable convolutions multiple times, the burden on the network is reduced while enhancing the fine-grained representation capability of features. The residual structure allows the encoder input features to be directly transmitted to the output, preserving the original details and combining them with deep semantic features, thereby reducing feature loss.
- (3) The CBAM attention is improved by constructing an Im-CSAM. The Im-CSAM consists of an improved channel attention mechanism (CAM) and an improved spatial attention mechanism (SAM). The Im-CSAM replaces the fully connected layers in traditional structures with depthwise separable convolutions, achieving more efficient feature extraction and enhancing the modeling capability of the network, thereby improving the CAM; it introduces the computation method of Query, Key, and Value, and uses multiple heads to compute attention independently, enhancing the SAM's ability to model local and global features.
- (4) The bicubic interpolation up-sampling method is used to restore image resolution and further improve the clarity of segmented images. Bicubic interpolation uses the pixel neighborhood data of 4×4 around the target pixel point to estimate the value of the target pixel by constructing a smooth interpolation function, thereby achieving high-quality results during image scaling.

2. Materials and Methods

In the cloud image segmentation process, shallow features have higher resolution and mainly represent information such as texture, position, and spectrum. Deep features have lower resolution and contain richer semantic information. Due to the strong feature extraction capability of CNNs, they are widely used in image segmentation tasks, such as fully convolutional networks (FCNs) [18], U-Net [14], SegNet [19], etc. This paper proposes a deep convolutional neural network based on an improved U-Net, which has an overall encoder–decoder structure. Its main architecture and key components are shown in Figure 1.

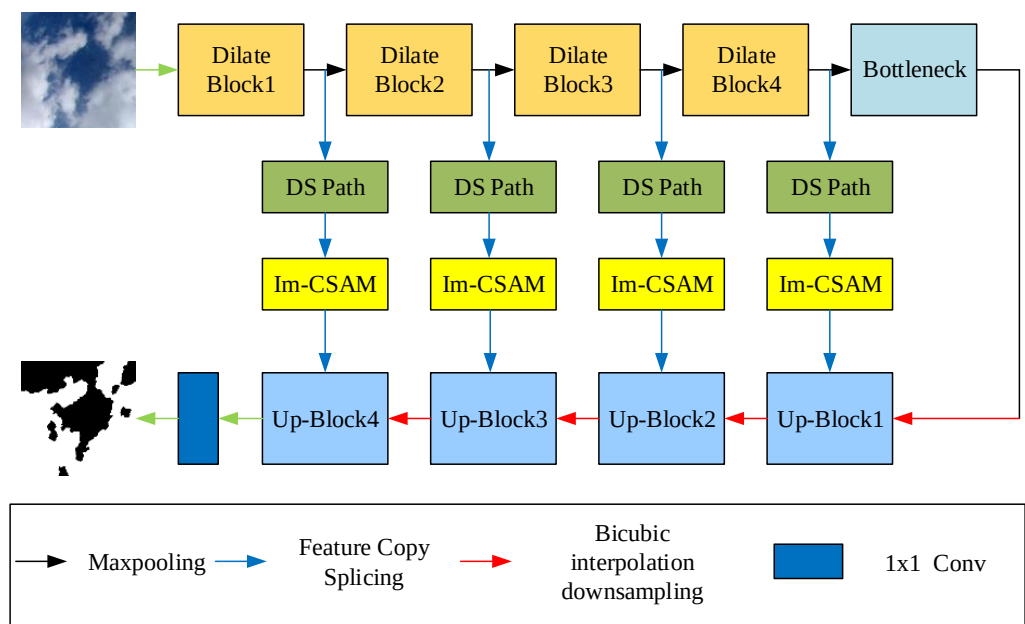


Figure 1. The overall architecture and key components of the model proposed in this paper.

2.1. Encoder

The designed encoder module in this paper consists of four dilate blocks and four down-sampling processes; each dilate block contains two dilated convolutions and one ASPP, as shown in Figure 2.

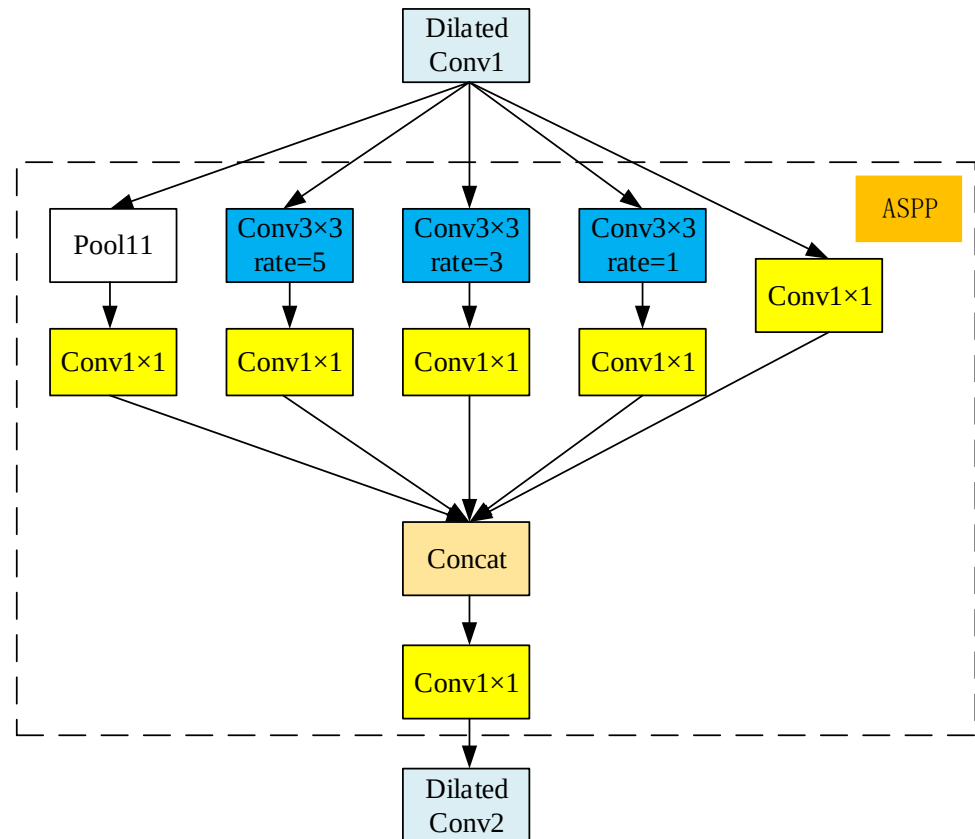


Figure 2. The main components of the dilate block.

Dilated convolution is a special type of convolution operation. In standard convolution, the elements of the convolution kernel are arranged consecutively, while in dilated convolution, gaps are inserted between the elements of the convolution kernel. This increases the receptive field of dilated convolution without directly increasing the size of the kernel, allowing it to capture features over a larger range, thereby avoiding the issue of information loss during down-sampling [20,21]. The cloud structure in the image spans multiple scales, including large-scale cloud clusters and local small cloud clusters. In order to further enhance the ability to extract cloud image features, this paper adds ASPP between the two dilated convolutions in each layer of dilate blocks. ASPP [20] was proposed to combine semantic information from different receptive fields without losing contextual information, thereby improving segmentation accuracy.

Therefore, this paper combines dilated convolution with ASPP, replacing traditional convolution layers with dilated convolution as the encoding module. This approach retains the fine structure of cloud images and reduces the loss of cloud layer information, while capturing a larger range of contextual information, completing the feature extraction of shallow cloud images. Additionally, by embedding ASPP between two layers of dilated convolution and setting convolution kernels at different dilation rates, multiple scales of cloud features are extracted in parallel. This allows the encoder to simultaneously capture large-scale background information and small-scale detail features, adapting to the complex multi-scale structure of cloud images.

The encoder accepts a $128 \times 128 \times 3$ RGB cloud image as input, and after feature extraction using four layers of dilate blocks, the output size is $8 \times 8 \times 128$. The bottleneck

layer, located at the bottom of the network, receives the output from the encoder and operates through two convolutional layers to effectively filter and compress the information, allowing the most important features to be retained and passed on to the decoder part.

2.2. Decoder

The goal of the decoder network is to obtain cloud mask output images of the same size as the input images to the encoder module. The decoding module in this paper consists of four up-blocks and a 1×1 convolution. Among them, each up-block module includes two convolutional layers with dimensions of 3×3 and adds batch normalization processing and activation layer functions. Advanced cloud feature alternates through up-sampling layers and up-block convolution blocks, finally obtaining a cloud mask of size 128×128 , with the same resolution as the input image.

Compared with non-pooling, up-sampling not only preserves the position information but also fills reasonable values during the interpolation process. Compared with deconvolution, up-sampling has better computational efficiency [22]. To improve the resolution of features, this paper uses bicubic interpolation to replace the traditional transposed convolution up-sampling method.

Bicubic interpolation is an interpolation method based on cubic polynomials. It considers the pixel values of a 4×4 grid surrounding the target pixel and calculates the interpolation result based on the grayscale values and positional relationship of the 16 points. Assume that $f(x, y)$ is the pixel value function of the input image. For the target point P , it is obtained by weighting and averaging the 4×4 sampling points in the rectangular grid, so the interpolation function is

$$P(x, y) = \sum_{i=0}^3 \sum_{j=0}^3 a_{ij} x^i y^j \quad (1)$$

where i and j represent the horizontal and vertical positions of the image respectively, and a_{ij} represent the weighting coefficient. The solution equation is as follows:

$$W(x) = \begin{cases} (a+2)|x|_3 - (a+3)|x|_2 + 1 & \text{for } |x| \leq 1 \\ a|x|_3 - 5a|x|_2 + 8a|x| - 4a & \text{for } 1 < |x| < 2 \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

where $a = -0.5$.

After the above calculation, $P(x, y)$ is the desired result of bicubic interpolation.

Therefore, the bicubic interpolation method considers the 4×4 sampling points around each target pixel in the cloud image. First, it calculates the weighted interpolation in the horizontal direction using cubic polynomials based on distance, and then, in the vertical direction, it calculates the final interpolation result using the weights based on the intermediate results obtained from the horizontal interpolation. The method is used to increase the image scale by increasing its resolution while minimizing the loss of image quality that occurs due to the enlargement of the image.

2.3. Skip Connection Layer Structure

The skip connection mechanism was first proposed in U-Net, which combines high-level semantic features from the decoder with low-level detailed features from the encoder, aiming to bridge the semantic gap between the encoder and decoder [23]. This paper proposes a novel skip connection layer structure, as shown in Figure 3, which consists of a DS path and an Im-CSAM connected in front and behind. The Im-CSAM is composed of an improved CAM and an improved SAM. The designed DS path combines the advantages of residual structures and depthwise separable convolutions, enhancing edge and texture details while improving across-layer feature fusion. In addition, the Im-CSAM has also improved the structure of channel and spatial attention in the CBAM, enhancing

and combining the CBAM's feature extraction capabilities in both spatial and channel dimensions.

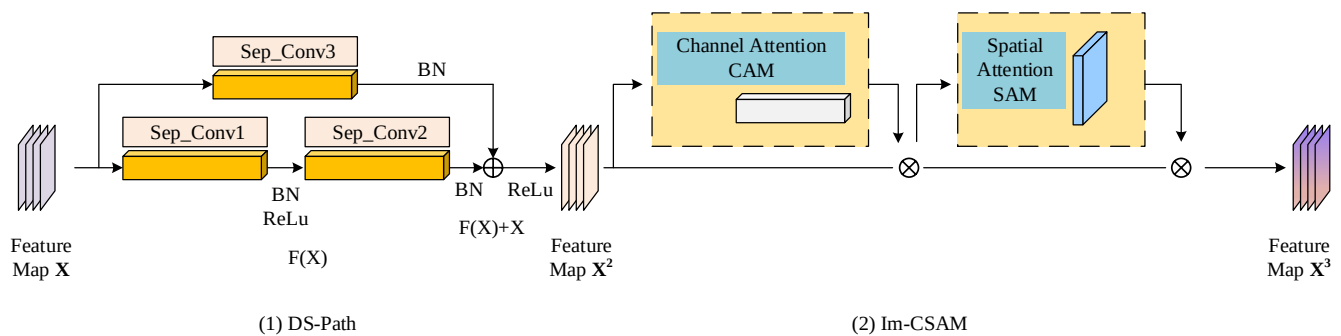


Figure 3. The main structures of the skip connection layer are (1) a DS path and (2) an Im-CSAM.

(1) DS path structure

ResNet excels in feature extraction and discrimination [24]. By utilizing the residual structure, the original input information can be directly transmitted to the output layer, mitigating the gradient vanishing issue associated with deep networks, and enhancing model performance [25]. The key to residual learning is to learn the residuals between inputs and outputs through the mapping layers. Each convolutional layer establishes shortcut connections between its input and output, allowing the input to be directly added to the output layer, forming the final target mapping. Therefore, the residual structure can achieve deeper architectures by reusing feature maps, thereby enhancing its learning capability.

Depthwise separable convolution is composed of depthwise convolution and pointwise convolution, which significantly reduces the number of model parameters and computation while maintaining better performance compared to standard convolution [26]. The difference between depthwise convolution and standard convolution lies in the fact that depthwise convolution has a single-channel kernel. The core idea is to perform independent convolution operations on each input channel without any interaction between different channels, resulting in an output feature map that has the same number of channels as the input feature map. However, this type of operation does not effectively utilize the feature information from different channels at the same spatial location. Therefore, pointwise convolution is needed to recombine these feature maps. Pointwise convolution is typically a 1×1 convolution operation that maps the features from multiple channels obtained after depthwise convolution to new channels through a weighted combination, ultimately outputting a new feature map.

As shown in Figure 3(1), the input X of the DS path originates from the low-level features of the dilate block. After the first depthwise separable convolution in each layer of the DS path, a batch normalization layer and ReLU activation function are applied. After the second depthwise separable convolution, only a batch normalization operation is performed. Its output is concatenated with the features from the shortcut and then passed through a ReLU layer to obtain feature map X^2 , which is input into the Im-CSAM module. In segmentation tasks, the boundaries and shapes of cloud layers are often complicated. The use of residual structures allows each layer's encoder output features to be effectively utilized and retained. Especially in the complex hierarchical structure of cloud images, residual structures can ensure the effective fusion of high-level and low-level features. Depthwise separable convolution replaces the standard convolution in traditional residual structures, where depthwise convolution retains the texture details of the cloud layers while pointwise convolution helps integrate information across channels, making cloud image segmentation more accurate and detailed.

(2) Im-CSAM structure

In the image segmentation task, not only the contextual information but also the spatial information should be paid attention to. A CBAM can combine information on two different features. A CBAM consists of channel attention and spatial attention, and its structure is shown in Figure 3(2).

For a traditional CBAM, assuming a tensor $F \in \mathbb{R}^{C \times H \times W}$ is given as input, the CBAM will sequentially derive a 1D channel attention feature map $M_c \in \mathbb{R}^{C \times 1 \times 1}$ and a 2D spatial attention feature map $M_s \in \mathbb{R}^{1 \times H \times W}$. The whole attention process can be summarized as

$$\begin{aligned} F' &= M_c(F) \otimes F \\ F'' &= M_s(F') \otimes F' \end{aligned} \quad (3)$$

Therefore, the final output feature map $RF \in \mathbb{R}^{C \times H \times W}$ can be obtained.

A CAM mainly focuses on discovering the key features in images. In a CAM, the importance of each channel is summarized through global average pooling and global max pooling, and then channel attention weights are generated through a shared MLP. Finally, these weights are used to re-weight each channel of the feature map. In short, the channel attention is calculated as follows [27]:

$$\begin{aligned} M_c(F) &= \sigma \{ \text{MLP}[\text{AvgPool}(F)] + \text{MLP}[\text{MaxPool}(F)] \} \\ &= \sigma \left\{ W_1 \left[W_0 \left(F_{\text{avg}}^c \right) \right] + W_1 \left[W_0 \left(F_{\text{max}}^c \right) \right] \right\} \end{aligned} \quad (4)$$

A SAM differs from a CAM but complements it, focusing more on identifying the key regions of important features. In a SAM, the importance of each location is summarized through global average pooling and global max pooling of the feature map along the channel dimension, and then spatial attention weights are generated through a convolutional layer. Finally, these weights are used to re-weight each position in the feature map. In short, the spatial attention is calculated as follows [27]:

$$\begin{aligned} M_s(F) &= \sigma \{ f^{7 \times 7} [\text{AvgPool}(F); \text{MaxPool}(F)] \} \\ &= \sigma \left[f^{7 \times 7} \left(F_{\text{avg}}^s, F_{\text{max}}^s \right) \right] \end{aligned} \quad (5)$$

To focus on the category and location information of clouds and their shadows, this paper replaces the MLP in the CAM with depthwise separable convolutions to focus on the important information and key features in the cloud image. At the same time, the SAM is combined with the multi-head attention mechanism to capture more spatial relationships in the feature map and find the key areas in the key features.

As shown in Figure 4, an Im-CSAM consists of an improved CAM and SAM. The Im-CSAM accepts the output X^2 from the previous module DS path.

In the improved CAM, the feature map X^2 is down-sampled by the max pooling layer and the average pooling layer respectively, and then enters the depthwise separable module with a two-layer structure. Its output is element-wise added, and the channel attention feature is generated through the sigmoid activation function. Finally, it is element-wise multiplied with the input feature X^2 to generate the input features required by the SAM module. Therefore, the improved channel attention is calculated as follows:

$$M_c(X^2) = \sigma \left\{ \text{DSConv} \left[\text{AvgPool}(X^2) \right] + \text{DSConv} \left[\text{MaxPool}(X^2) \right] \right\} \quad (6)$$

where DSConv represents the depthwise separable convolution operation.

Since the depthwise separable convolution contains depthwise convolution and point-wise convolution, Equation (6) can be expanded as follows:

$$\mathbf{M}_c(\mathbf{X}^2) = \sigma \left\{ \mathbf{W}_{p1} \left[\mathbf{W}_{d1} \left(\mathbf{X}_{\text{avg}}^{2c} \right) \right] + \mathbf{W}_{p2} \left[\mathbf{W}_{d2} \left(\mathbf{X}_{\text{max}}^{2c} \right) \right] \right\} \quad (7)$$

where $\mathbf{W}_{d1}$ and $\mathbf{W}_{d2}$ are the weights of the depthwise convolution, $\mathbf{W}_{p1}$ and $\mathbf{W}_{p2}$ are the weights of the pointwise convolution, and $\mathbf{X}_{\text{avg}}^{2c} = \text{AvgPool}(\mathbf{X}^2)$ and $\mathbf{X}_{\text{max}}^{2c} = \text{MaxPool}(\mathbf{X}^2)$ represent the results of global average pooling and global max pooling, respectively.

In the SAM, based on Equation (5), multi-head self-attention is used to replace the original global average pooling and global max pooling operations, which accepts features from the CAM as input, and the improved spatial attention is calculated as follows:

$$\mathbf{M}_s(\mathbf{X}^{2c}) = \sigma \left[\text{MultiHeadAttention}(\mathbf{X}^{2c}) \right] \quad (8)$$

where MultiHeadAttention represents the multi-head self-attention mechanism.

Specifically, the multi-head self-attention is composed of N single-head self-attention components, where the feature vectors are mapped to K_n (key vectors), Q_n (query vector), and V_n (value vector), where n represents the number of self-attention mechanism heads. After K_n and Q_n undergo the Matmul operation (denoted as M) to produce a new matrix, the scaled dot-product operation (denoted as S) is applied, followed by another M operation with V_n to obtain the attention score matrix [28,29], and then, the attention score and weighted sum value are calculated independently for each head, the outputs of the N heads are concatenated together, and finally linearly transformed through the forward propagation network to obtain the final output feature $\mathbf{X}^3$.

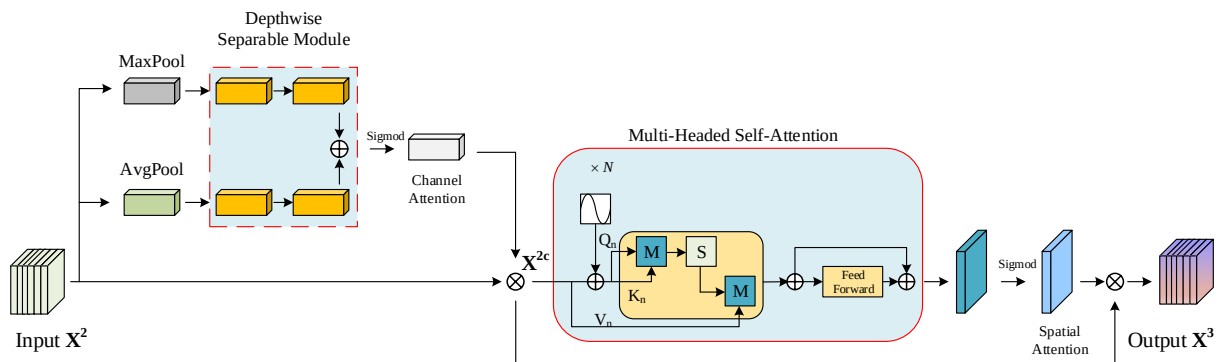


Figure 4. Main components of optimized CAM and SAM in Im-CSAM.

The improved U-Net model integrates the strengths of U-Net, novel convolutions, ASPP, and multi-head attention, boosting the network's feature extraction capabilities. This enables the network to capture high-level semantic information while paying attention to details, thereby enhancing target extraction capabilities.

3. Experimentation and Results

This section provides a detailed description of the required dataset, evaluation metrics, and comparison results. This paper is based on the win11 operating system and uses the Python 3.9 programming language and Tensorflow as the deep learning framework. It compares the improved U-Net with existing mature image segmentation models such as U-Net, FCN, SegNet, and CloudSegNet.

3.1. Experimental Environment and Dataset

This experiment was designed to be conducted on a high-performance server platform equipped with an Intel (R) Core (TM) i9-12900KF CPU, 32 GB RAM (Intel, Santa Clara, CA, USA), and NVIDIA GeForce RTX3080ti (Nvidia, Santa Clara, CA, USA). The experiment used the SWINySEG dataset made publicly available by the National University of Singapore [15] and the TCDD dataset made publicly available by Tianjin Normal University [22].

The SWINySEG dataset includes 6768 daytime and nighttime cloud images along with their corresponding binary label maps, while the TCDD dataset consists of 1874 ground cloud images and their corresponding cloud masks. In this experiment, both datasets were divided, with 90% used for the training set and 10% for the test set.

3.2. Evaluation Indicators

In this paper, the six most common evaluation metrics in the field of image segmentation are selected to measure the overall performance of the segmentation model, which are accuracy (A), precision (Pr), recall (Re), F1 score (F1-score), mean intersection over union (MIoU), and error rate (ER), with the following equation:

$$A = \frac{TN + TP}{TP + FP + TN + FN} \quad (9)$$

$$Pr = \frac{TP}{TP + FP} \quad (10)$$

$$Re = \frac{TP}{TP + FN} \quad (11)$$

$$F1 = 2 \times \frac{Pr \times Re}{Pr + Re} \quad (12)$$

$$MIoU = \frac{1}{2} \left(\frac{TP}{TP + FP + FN} + \frac{TN}{TN + FP + FN} \right) \quad (13)$$

$$V_{ER} = \frac{FP + FN}{TP + TN + FP + FN} \quad (14)$$

where TP is the number of positive samples that are correctly classified; TN is the number of negative samples that are correctly classified; FN is the number of positive samples that are misclassified; FP is the number of negative samples that were misclassified.

In the experiments conducted for this paper, accuracy is directly related to the segmentation results; a higher accuracy indicates that most pixels are classified correctly. A higher recall rate means that the segmentation model does not miss many cloud pixels; a higher precision means that the areas predicted as clouds are less often misclassified as non-cloud areas. The F1 score considers both precision and recall, reflecting the balance between the precision and recall capabilities of the model. The MIoU focuses more on the accurate segmentation of cloud areas and background areas. The better the segmentation results, the higher the MIoU value, indicating the model's more precise identification of cloud areas. A higher error rate means that there are more misclassifications in the model's segmentation results, especially in the segmentation of clouds and backgrounds.

3.3. Example Analysis

3.3.1. Ablation Experiment

In order to verify the effectiveness of the dilate block, DS path, and Im-CSAM proposed in this paper for segmentation, ablation experiments were conducted on the SWINySEG and TCDD datasets. In the experiment, the dilate block, DS path, and Im-CSAM were removed separately, and the ablation experiment results were analyzed using three evaluation metrics: accuracy, MIoU, and ER.

Table 2 shows the ablation experiment results on two datasets. Taking the SWINySEG dataset as an example, the basic model in this paper (referred to as Model a) has an accuracy of 0.876, an MIoU of 0.763, and an ER of 0.124 during model training. Adding dilate blocks to Model a (now referred to as Model b) increases accuracy by 0.002 and the MIoU by 0.021, and decreases the ER by 0.002. Adding a DS path to Model a (now referred to as Model c) increases accuracy by 0.008 and the MIoU by 0.029, and decreases the ER by 0.008. Adding an Im-CSAM to Model a (now referred to as Model d) increases accuracy by 0.01 and the MIoU by 0.033, and decreases the ER by 0.01. Compared to Model a, the accuracy of the proposed model is increased by 0.019, the MIoU is increased by 0.038, and the ER is

decreased by 0.019. Both the accuracy and MIoU are higher than in Model b, Model c, and Model d.

Similarly, on the TCDD dataset, the performance metrics of Models b, c, and d all surpass those of the baseline Model a, while the proposed model in this paper exhibits the best performance. Compared to Model a, accuracy increases by 0.029, the MIoU increases by 0.049, and the ER decreases by 0.028. Compared to Models b, c, and d, all metric values show stable improvements, indicating that the proposed dilate blocks, DS path, and Im-CSAM all enhance the segmentation performance of the model.

Table 2. The results of the ablation experiment on the SWINySEG and TCDD datasets.

Model	Dilate Block	DS Path	Im-CSAM	A	MIoU	ER
SWINySEG Dataset						
Model a				0.876	0.763	0.124
Model b	√			0.878	0.784	0.122
Model c		√		0.884	0.792	0.116
Model d			√	0.886	0.796	0.114
Improved U-Net	√	√	√	0.895	0.801	0.105
TCDD Dataset						
Model a				0.914	0.844	0.085
Model b	√			0.921	0.848	0.078
Model c		√		0.933	0.875	0.066
Model d			√	0.931	0.880	0.069
Improved U-Net	√	√	√	0.943	0.893	0.057

3.3.2. Comparison of Different Model Indicators

Table 3 shows the evaluation results of the improved U-Net in this paper and four other segmentation network models on the cloud image datasets.

Table 3. A comparison of the evaluation indicators of different segmentation networks on the SWINySEG and TCDD datasets.

Models	A	Pr	Re	F1-Score	MIoU	ER
SWINySEG Dataset						
FCN	0.848	0.863	0.807	0.834	0.715	0.152
U-Net	0.876	0.885	0.847	0.866	0.763	0.124
SegNet	0.855	0.890	0.847	0.868	0.767	0.145
CloudSegNet	0.869	0.862	0.862	0.862	0.758	0.130
Improved U-Net	0.895	0.921	0.850	0.884	0.801	0.105
TCDD Dataset						
FCN	0.916	0.911	0.910	0.915	0.843	0.084
U-Net	0.914	0.922	0.931	0.915	0.844	0.085
SegNet	0.899	0.899	0.898	0.899	0.816	0.101
CloudSegNet	0.933	0.921	0.926	0.920	0.852	0.067
Improved U-Net	0.943	0.955	0.970	0.944	0.893	0.057

According to Table 3, taking the SWINySEG dataset as an example, in the comparison of accuracy metrics, the accuracy of the FCN is 0.848, followed by SegNet at 0.855, both with

relatively low accuracy. The proposed improved U-Net model has the highest accuracy of 0.895, with the overall segmentation accuracy being the highest.

In the comparison of the metrics of precision, CloudSegNet and FCN have similar precision rates and also maintain the lowest value, U-Net and SegNet have precision rates of 0.885 and 0.890, respectively, and the proposed model in this paper has the highest precision rate of 0.921, which represents the highest level of segmentation precision among the results of positive samples.

In the comparison of the metrics of recall, the FCN has the worst performance of only 0.807, U-Net and SegNet have the same performance, and the recall value of the improved U-Net in this paper is only second to CloudSegNet, with a difference of only 0.012, which is not a significant improvement, but it reaps better results.

The F1-score is a comprehensive evaluation index, and the highest F1-score of 0.884 for the improved U-Net is higher than the F1-scores of the other four models, with the lowest F1-score of 0.834 for the FCN.

The MIoU is a relatively important evaluation indicator in semantic segmentation tasks. The lowest MIoU value of the FCN is 0.715, followed by CloudSegNet with a MIoU value of 0.758. The performance of U-Net and SegNet is relatively close. Compared with the other four models, the improved U-Net has improved by up to 12.03%, proving that it has a larger overlapping area with the true label during segmentation prediction and has the best segmentation effect.

The ER index is an extension of the accuracy index. Only the improved U-Net proposed in this paper performs best, with the lowest ER index of 0.105, and the highest ER index of the FCN is 0.152. Combined with the accuracy index, the FCN has the lowest segmentation accuracy and the highest error rate.

Similarly, taking the TCDD dataset as an example, SegNet's accuracy, precision, recall, and F1-score all remain at a low level. In comparisons of accuracy, F1-score, MIoU, and ER, U-Net and the FCN perform similarly, with differences fluctuating between 0 and 0.002. CloudSegNet shows a slight improvement on TCDD. The six evaluation metrics for the proposed improved U-Net are 0.943, 0.955, 0.970, 0.944, 0.893, and 0.507, indicating that its performance on the TCDD dataset is also the best, with the most stable improvement.

Combining the above six metrics, five of the six evaluation metrics of the improved U-Net are higher for the SWINySEG dataset than for the other models, and the recall value is just 0.012 lower than the best-performing CloudSegNet. For the TCDD dataset, the improved U-Net evaluation metrics are superior to those of other models. The experimental comparison of different metrics proves the superior performance and best segmentation results of the improved U-Net proposed in this paper in the cloud image segmentation task.

3.3.3. Comparison of Experimental Segmentation Results

In order to further prove the superiority of this paper's model in cloud image segmentation tasks, segmentation experiments were conducted using the FCN, U-Net, SegNet, CloudSegNet, and the improved U-Net, with the results shown in Figures 5 and 6.

Figures 5 and 6 show ten test cloud images along with their corresponding segmentation binary maps representing the FCN, U-Net, SegNet, CloudSegNet, and improved U-Net. Black represents the sky while white represents the cloud layer. Figure 5a–e are from the SWINySEG dataset, and Figure 6a–e are from the TCDD dataset.

Taking the SWINySEG dataset as an example, Figure 5a shows that SegNet exhibits the poorest segmentation performance. Because it fails to clearly delineate the boundaries between the clouds and the sky, it leads to numerous missed detections, with much of the sky being erroneously classified as clouds. CloudSegNet can roughly identify the general area and distribution of the clouds and the sky, but cannot accurately describe and segment the details of cloud edges. The segmentation results of traditional U-Net and the FCN are relatively good, showing a significant improvement compared to SegNet, as they can capture more shape contour information of cloud bodies, and the contour information of cloud layer edges has also been further optimized. However, for the clouds closest to the

edge of the image, the segmentation effect of traditional U-Net is poor, making it difficult to distinguish between the sky and the clouds, and the FCN also struggles to achieve a more detailed edge depiction. Figure 5c shows that U-Net and SegNet have rough processing of boundaries, and the features of small cloud patches are not retained. Although the FCN and CloudSegNet can roughly depict the texture details of cloud images, their ability to distinguish between blue sky and small cloud patches is insufficient. The improved U-Net has the best overall performance, with rich segmentation details and clear boundaries, capable of capturing the complex shapes of clouds, and its segmentation results are more aligned with the real cloud patch shapes.

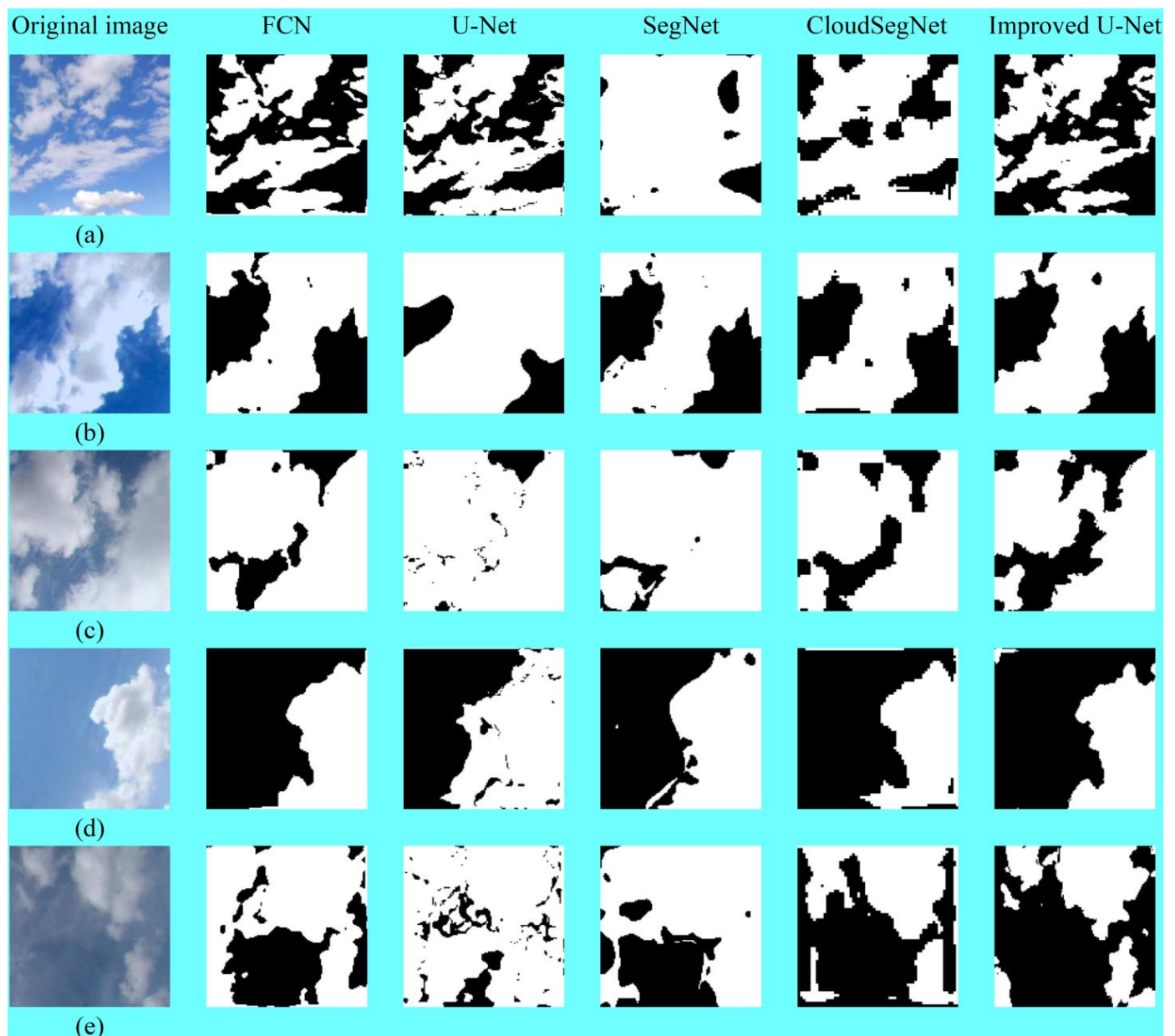


Figure 5. A comparison of segmentation results based on the SWINySEG dataset. (a–e) represents five sets of test images and their segmentation results under different models.

In light of the TCDD dataset (Figure 6a–e), the segmentation results produced by the FCN are rather crude, with blurry and inaccurate cloud boundaries. The FCN is suitable for preliminary segmentation, but the preservation of details is poor, leading to missed detections. In comparison to the FCN, U-Net exhibits improved segmentation

accuracy, successfully capturing the majority of the cloud's primary regions, albeit with some boundary details missing. Additionally, its performance on small cloud blocks is not as satisfactory. Given that SegNet inherently possesses a lightweight characteristic with fewer model parameters, its segmentation results exhibit significant noise, thus being unable to accurately depict cloud formations. CloudSegNet is an improved version of SegNet, thereby enabling a more accurate depiction of cloud structures. Yet, CloudSegNet does not use fully connected layers and abandons skip connections, resulting in a significant reduction in the number of trainable parameters. Consequently, the segmentation accuracy improvement is limited. The improved U-Net shows the best performance, significantly enhancing its ability to capture small cloud blocks. The segmentation results indicate that the boundaries are clearer, and the shapes and sizes of the clouds are more accurate, with better processing capabilities for background noise compared to other networks.

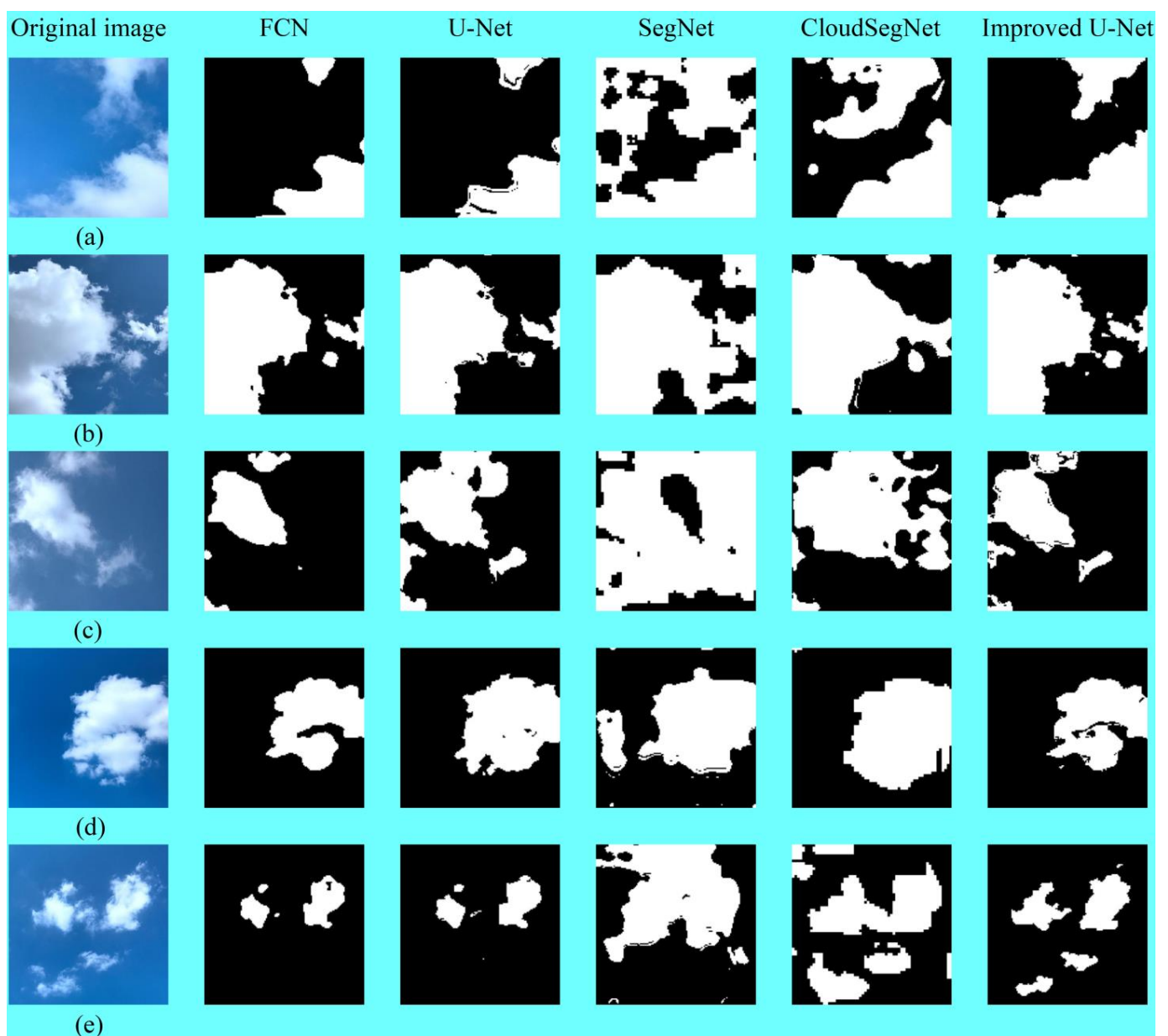


Figure 6. A comparison of segmentation results based on the TCDD dataset. (a–e) represents five sets of test images and their segmentation results under different models.

Combining Table 3 and Figures 5 and 6, the FCN and SegNet performed poorly, struggling to handle the details and noise in complex cloud images. The traditional U-Net and CloudSegNet can capture the main parts of the clouds, but they lack the ability to segment cloud details. The improved U-Net performed best in segmentation tasks, accurately capturing the edges and shapes of the clouds while reducing false positives and negatives, successfully achieving more precise segmentation results.

4. Conclusions

This paper designs and constructs a deep learning model for cloud image segmentation, which consists of three parts: an encoder, a decoder, and a skip connection layer. The skip connection layer embeds the depthwise separable residual block DS path and the optimized Im-CSAM. Compared to traditional segmentation networks, firstly, the improved U-Net encoder combines the advantages of dilated convolutions and ASPP, preserving the fine features of the input cloud image and reducing the loss of key information in the cloud layers. Secondly, this paper constructs a residual block centered on depthwise separable convolutions and combines depthwise separable convolutions with multi-head attention to optimize channel attention and spatial attention. This is embedded in a cascading manner into the skip connection layers, which helps retain original details while integrating with deep semantic features, reducing feature loss, and significantly enhancing the improved U-Net's ability to extract important information and key features in both spatial and channel dimensions. Finally, in the decoder section, a bicubic interpolation algorithm is used to achieve high-quality segmentation results during the image resolution recovery process. The improved U-Net proposed in this paper achieved optimal segmentation metrics on both the SWINySEG and TCDD datasets. Ablation experiments show that the dilate block, DS path, and Im-CSAM all improve the segmentation performance. In conclusion, the improved U-Net has the best performance and segmentation effect, which is of consequence for cloud amount calculation, weather forecasting, PV power prediction, and other fields. However, the parameter size of the improved model is relatively large, and future work will focus on further optimizing the network structure and reducing the parameter size of cloud image segmentation through lightweight methods.

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